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Case 5-2024: A 36-Year-Old Man with Fevers

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PRESENTATION OF CASE

Dr. Ethan Borre (Medicine): A 36-year-old man was evaluated in the infectious diseases clinic of this hospital because of fevers.

The patient had been in his usual state of health until 18 days before the current presentation, when he began to have a sensation of swelling in his throat. Three days later, the temporal temperature measured at home was 38.4°C. A home test for severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) antigen was negative. The patient was evaluated in a local urgent care clinic, where he was told that there were white spots on his tonsils. Treatment with amoxicillin–clavulanic acid was started for a presumptive diagnosis of streptococcal pharyngitis. A throat culture was reportedly negative, but treatment with amoxicillin–clavulanic acid was continued. During the next 4 days, fevers persisted. A repeat home test for SARS-CoV-2 antigen was negative. Eleven days before the current presentation, the patient was evaluated by his primary care physician.

In addition to fevers, the patient reported headaches. Two weeks before the onset of symptoms, he had traveled with his wife to the southwestern United States for 1 week. During the trip, his wife had begun to have fever and sore throat; her symptoms had resolved after several days, and she did not seek medical evaluation. The patient reported that he did not have throat pain, but rather that the lower part of his neck felt swollen and that moving his neck caused discomfort. He had no dysphagia, odynophagia, rhinorrhea, or rash. His medical history included seasonal allergies.

Two years earlier, a routine screening test for human immunodeficiency virus (HIV) had been negative. In addition to amoxicillin–clavulanic acid, medications included ibuprofen as needed for headaches and acetaminophen as needed for fevers. The patient had no known drug allergies. He lived in Massachusetts and worked in an office setting. He had moved to the United States from Asia 7 years before the current presentation. He drank alcohol rarely and did not use illicit drugs. He occasionally smoked tobacco. His father had hypertension, and his paternal grandmother had cancer of unclear primary origin.

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N Engl J Med 2024;390:653-60.

DOI: 10.1056/NEJMcpc2312724

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Table 1. Laboratory Data.

Variable	Reference Range*	11 Days before the Current Presentation
Hematocrit (%)	41.0–53.0	43.8
Hemoglobin (g/dl)	13.5–17.5	14.6
White-cell count (per μ l)	4500–11,000	7320
Differential count (per μ l)		
Neutrophils	1800–7700	5100
Lymphocytes	1000–4800	1430
Monocytes	200–1200	610
Eosinophils	0–900	130
Basophils	0–300	30
Immature granulocytes	0–100	20
Platelet count (per μ l)	150,000–400,000	375,000

* Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

On examination, the temporal temperature was 36.2°C, the blood pressure 127/69 mm Hg, the pulse 90 beats per minute, the respiratory rate 18 breaths per minute, and the oxygen saturation 97% while the patient was breathing ambient air. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 25.8. The tonsils appeared normal, without exudates or enlargement. There was mild tenderness on palpation over the anterior portion of the neck that was thought to correspond to enlarged lymph nodes in the anterior cervical region. The remainder of the examination was normal.

A complete blood count with differential count and blood levels of electrolytes, glucose, alanine aminotransferase, aspartate aminotransferase, alkaline phosphatase, and bilirubin were normal, as were the results of tests of kidney function. The blood level of C-reactive protein (CRP) was 65.8 mg per liter (reference value, <8.0). Other laboratory test results are shown in Table 1. Testing for Epstein–Barr virus (EBV) antibodies was negative. Urinalysis was normal. A specimen of blood was obtained for culture. The results of chest radiography were normal. A diagnosis of viral pharyngitis was considered, but treatment with amoxicillin–clavulanic acid for possible streptococcal pharyngitis was continued.

During the next 6 days, daily fevers and head-

aches continued, and fatigue developed. The patient reported his symptoms during a telephone call with his primary care physician. Blood cultures that had been obtained during the previous evaluation were without growth, and additional testing was performed. The blood level of CRP was 138.9 mg per liter, and the erythrocyte sedimentation rate (ESR) was 81 mm per hour (reference range, 0 to 14). A test for Lyme disease was nonreactive. The result of an interferon- γ release assay to detect previous exposure to *Mycobacterium tuberculosis* was invalid because of a test control failure. Imaging studies were obtained.

Dr. Shambo Guha Roy: Computed tomography (CT) of the neck (Fig. 1), chest, and abdomen was performed after the administration of intravenous contrast material. A calcified lung nodule that was observed in the left lower lobe was consistent with previous granulomatous disease, and mild degenerative changes were noted in the thoracic spine. Evaluation of the thyroid gland was suboptimal owing to a streak artifact from the arms; however, a possible hypoattenuating nodule was observed in the right lobe.

Dr. Borre: During the next 4 days, fevers continued, and the patient was referred to the infectious diseases clinic of this hospital for further evaluation.

In the infectious diseases clinic, additional history was obtained. The patient reported that fevers were most prominent in the evening; the highest recorded temperature was 39.5°C. He had had two episodes of chills associated with fever, as well as occasional night sweats. He had no pets or animal exposures. He did not consume unpasteurized cheeses or uncooked meats. He was sexually active with his wife. A dental extraction had been performed 2 months before the onset of fevers. He had no known exposures to persons with tuberculosis.

The temporal temperature was 36.6°C, the blood pressure 144/98 mm Hg, the pulse 130 beats per minute, the respiratory rate 18 breaths per minute, and the oxygen saturation 99% while the patient was breathing ambient air. He was noted to cough occasionally during the examination but was breathing comfortably with normal lung sounds. The oropharynx appeared normal, without tonsillar exudates. There was no supraclavicular, submandibular, cervical, axillary, or inguinal lymphadenopathy. Firm, nontender thyroid tissue was palpable in the anterior por-

tion of the neck. No rashes or skin lesions were present, including on the palms or soles. The heart was tachycardic with a regular rhythm and no murmurs. There was no hepatosplenomegaly.

A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Eric A. Meyerowitz: This 36-year-old man presented with subacute fevers, along with 18 days of associated symptoms, including throat discomfort, headaches, and fatigue. He had an appropri-

ately extensive but unrevealing workup. Given that this patient's illness was characterized by fevers despite the administration of amoxicillin-clavulanic acid, I will first focus on some common noninfectious causes of fever and will then consider infectious and infection-associated causes using a systematic approach in the context of this host, his exposures, the time course of his illness, and the clinical syndrome.

NONINFECTIOUS CAUSES

Medication-induced fever is common but is unlikely in the case of this patient, since his fevers predated the administration of any medications. The patient reported occasional night sweats, but he had no weight loss and no other risk factors that would suggest that the fevers were due to an underlying cancer. He had no other signs or symptoms that were consistent with rheumatologic disease, which makes an autoimmune process unlikely.

HOST IMMUNE STATUS

The clinical presentation and risk of various infections depend on the nature of the host's immune status. Immunocompetent adults are likely to have typical presentations of common or unusual infections, whereas immunocompromised persons may have typical or atypical presentations of common or unusual infections.

Defining specific immune defects can narrow a differential diagnosis involving infection. For instance, diabetes confers a risk of invasive group B streptococcal infections, and eczema can confer a predisposition to severe manifestations of herpes simplex virus (HSV) infection and mpox. Certain intrinsic or acquired immunodeficiencies (such as HIV infection, exposure to certain medications, or previous organ transplantation) may be associated with a wide range of infections, including opportunistic infections. This patient had a negative routine screening test for HIV approximately 2 years before the current presentation, which makes severe immunodeficiency from longstanding, untreated HIV infection unlikely. Although severe CD4⁺ T-cell depletion can occur during acute HIV infection and is occasionally associated with opportunistic infections, this patient had no identified risk factors for a new HIV infection. Therefore, for the purpose of narrowing the list of possible diagnoses, I will presume that this patient is immunocompetent.

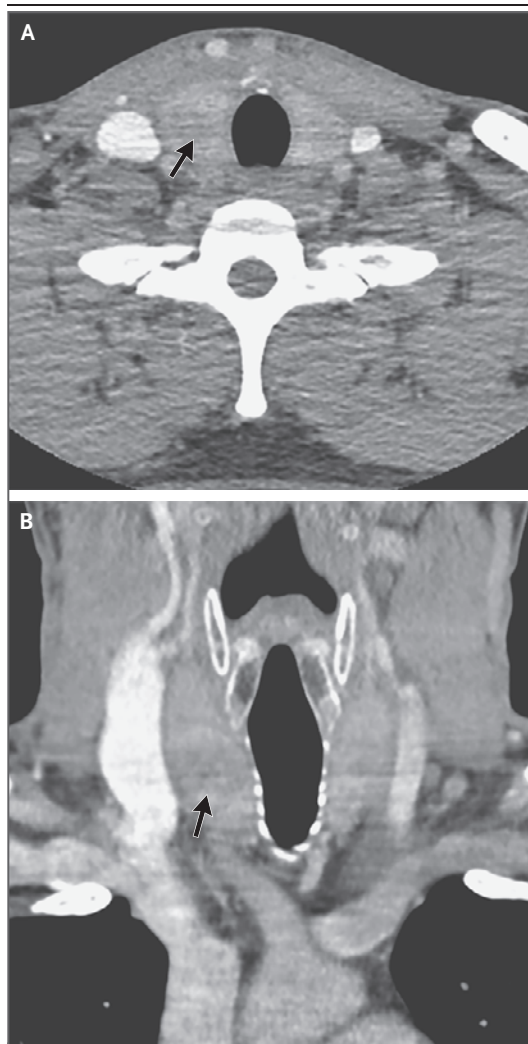


Figure 1. CT of the Neck.

Axial (Panel A) and coronal (Panel B) CT images of the neck were obtained after the administration of intravenous contrast material. The thyroid gland has a heterogeneous appearance, and a possible nodule is seen in the right lobe (arrows).

EXPOSURE TO INFECTION

Exposure to infection is important to consider but only in the context of a consistent clinical syndrome. For instance, this patient had recently traveled to the southwestern United States, where coccidioidomycosis is an important consideration. However, most persons with primary coccidioidomycosis remain asymptomatic. Approximately one third of persons with coccidioidomycosis have valley fever (an acute fungal pneumonia)¹; however, this patient had no cough and no opacities on chest imaging, which makes primary coccidioidomycosis unlikely to explain his fevers.

The subacute onset, time course, and stability of the illness should also be considered. This patient had no signs of organ dysfunction, and his condition remained hemodynamically stable throughout the course of his illness. These features rule out rapidly progressive processes such as septic thrombophlebitis of the internal jugular vein.²

DEFINING THE CLINICAL SYNDROME

The patient described his first symptom as a sensation of throat swelling. A few days later, an urgent care provider reported “white spots” on his tonsils, which suggested a possible clinical syndrome of exudative pharyngitis and led to a presumed diagnosis of streptococcal pharyngitis. Tonsillar exudates are nonspecific and may be associated with both bacterial and viral processes.³ Posterior oropharyngeal or tonsillar vesicles resulting from certain viral infections, such as HSV infection, could also appear as white spots.⁴ Exudative pharyngitis typically manifests with pain during swallowing and tenderness on palpation of enlarged lymph nodes.^{5,6} When this patient was evaluated by his primary care physician, he clarified that he was having a sensation of swelling in the lower portion of his neck rather than a sore throat or pain or difficulty with swallowing, and no tonsillar exudates were seen. The primary care physician noted anterior neck tenderness that was possibly due to enlarged anterior cervical lymph nodes; however, a CT image obtained a few days later did not show lymphadenopathy in the neck.

Eleven days later, during the current evaluation in the infectious diseases clinic, the heart rate was 130 beats per minute, there was no fever, and palpable thyroid tissue was noted in the anterior portion of the neck. Although the thyroid

tissue was nontender on palpation in the clinic, it is likely that the anterior neck tenderness previously attributed to lymphadenopathy was pain on palpation of thyroid tissue. Taken together, the patient’s fevers, anterior neck swelling and tenderness over the thyroid gland, elevated levels of markers of inflammation, and sinus tachycardia could be indicative of hyperthyroidism. Although the patient was initially treated for exudative pharyngitis, the clinical syndrome is most consistent with subacute thyroiditis.

SUBACUTE THYROIDITIS

What could cause subacute thyroiditis in this patient? Subacute thyroiditis is associated with numerous preceding viral triggers, although the exact trigger is often unknown.⁷

Vaccine-Preventable Viruses

Several vaccine-preventable viral infections are associated with subacute thyroiditis, including measles, mumps, and rubella. Although this patient may have come from an undervaccinated region, the measles–mumps–rubella vaccine was required by the Department of Health and Human Services at the time of the patient’s immigration.⁸ No preceding clinical syndrome consistent with acute infection from these viruses was reported, such as a viral exanthem, conjunctival changes, or pain or swelling of the parotid gland or testicles.^{9,10}

Viruses Associated with Exposure to Children

Exposure to children in congregate settings, such as day care centers or schools, is a risk factor for exposure to viruses associated with subacute thyroiditis, including echovirus, parvovirus B19, coxsackievirus, and varicella–zoster virus (VZV). This patient had no reported high-risk exposure and no rashes, consistent sores, or arthralgias.^{11,12} He may have had primary VZV infection as a child and was unlikely to have had VZV vaccination during childhood on the basis of his age.¹³ Without previous infection, he would be susceptible to primary VZV infection¹⁴; however, no typical symptoms were described.

Viruses Associated with Sexual and Other Close Intimate Contact

A third group of viruses that are associated with the development of subacute thyroiditis are acquired through sexual and other close intimate

contact. These include cytomegalovirus (CMV), HSV types 1 and 2, EBV, and HIV. Typically, only recently acquired infections trigger subacute thyroiditis. It is important to ask about extramarital partners and to perform appropriate testing. A new HSV type 2 infection would not necessarily result from an additional sexual partner, since the baseline incidence of HSV type 2 transmission among serodiscordant couples is approximately 5% per year.^{15,16} However, this patient had no symptoms consistent with oral or genital HSV infection.

EBV, CMV, and HIV can all cause a mononucleosis syndrome with fever. However, pharyngitis, lymphadenopathy, and atypical lymphocytosis would also be expected.¹⁷ This patient's EBV serologic tests rule out acute or previous EBV infection; after nearly 3 weeks of symptoms, the EBV-specific antibodies would be detectable in a person with acute EBV infection.¹⁸ A negative test for HIV antigen and antibody would rule out acute HIV infection at this point in the patient's course of illness.¹⁹

One report of subacute thyroiditis that occurred after mpox virus infection described a 51-year-old man with well-controlled HIV infection who began to have neck discomfort, fatigue, weight loss, elevated CRP levels, and hyperthyroidism 3 weeks after infection.²⁰ Even though mpox can be asymptomatic, it seems highly unlikely in this case, in the absence of typical symptoms, rash, or additional sexual partners.

Respiratory Viruses

Infection with respiratory viruses, such as influenza virus, can also cause subacute thyroiditis. This patient had no history of an influenza-like illness; however, approximately 30% of cases of influenza are asymptomatic.²¹ Since the presumed preceding infection probably occurred several weeks before the current presentation, it may not be possible to confirm a remote influenza virus infection. However, the Centers for Disease Control and Prevention reports influenza case counts at all times; such information should be checked against the patient's time course of illness.²²

Several case reports and series describe the development of subacute thyroiditis after SARS-CoV-2 infection.²³ This patient's wife had had a

transient febrile illness with a sore throat — very common symptoms associated with multiple sublineages of the SARS-CoV-2 omicron (B.1.1.529) variant²⁴ — during their trip to the southwestern United States, approximately 2 weeks before the onset of swelling and tenderness over the patient's thyroid gland. The patient did not have similar symptoms at that time; however, approximately 20 to 30% of SARS-CoV-2 infections are asymptomatic.²⁵

The patient's two negative SARS-CoV-2 rapid antigen tests are not surprising, since the typical onset of subacute thyroiditis is at least 2 to 4 weeks after the inciting infection. Rapid antigen tests are positive when viral loads are highest, and less than 1% of tests are positive more than 10 days after infection.^{26,27} The vast majority of nucleic acid tests in immunocompetent young adults are negative 15 days after infection, and this patient's test would probably have been negative by the time of the current presentation.²⁸

I suspect that this patient had subacute thyroiditis, and the diagnostic test was measurement of the thyrotropin level. Serologic testing might confirm a recent SARS-CoV-2 infection, but it is not always reliable. Among unvaccinated persons with a new SARS-CoV-2 infection confirmed by a polymerase-chain-reaction test, more than 90% will subsequently have antibodies to the nucleocapsid protein.²⁹ By contrast, among previously vaccinated persons, only approximately 40% may subsequently have antinucleocapsid antibodies. Therefore, if this patient had been previously vaccinated, a negative test for antinucleocapsid antibodies would not rule out a recent infection. Antinucleocapsid antibody levels rise for approximately 100 days and then decline, with the incidence of seroreversion expected to be roughly 20% after 2 years.^{30,31}

CLINICAL IMPRESSION

Dr. Anne M. Neilan: This healthy 36-year-old man had nightly fevers with a temperature of up to 39.5°C over a period of 2.5 weeks. His fevers were preceded by a sensation of fullness in his neck. He had fatigue with no specific localizing symptoms. His examination was notable for a subtle finding of firm, nontender tissue in the anterior portion of the neck that was possibly

consistent with an enlarged thyroid gland. Given that the patient's previous extensive evaluation — which included blood cultures, laboratory tests, and CT of the abdomen, chest, and neck — was unrevealing, I thought that the most likely explanation for his fevers, night sweats, tachycardia, and fatigue was hyperthyroidism, and I obtained a thyroid panel.

culated by comparing the number of light units produced in the cell lysate on exposure to a patient's serum with that produced on exposure to a control serum. This patient's TSI index was less than 1.0 (negative; reference value, ≤ 1.3).

Overall, this patient's laboratory findings were consistent with hyperthyroidism. No anti-thyroid antibodies were detected.

CLINICAL DIAGNOSIS

Fevers most likely due to hyperthyroidism.

LABORATORY DIAGNOSIS

Hyperthyroidism.

DR. ERIC A. MEYEROWITZ'S DIAGNOSIS

Postviral subacute thyroiditis, possibly after a subclinical or asymptomatic severe acute respiratory syndrome coronavirus 2 infection.

DISCUSSION OF MANAGEMENT

Dr. Douglas S. Ross: Subacute granulomatous thyroiditis, commonly called subacute thyroiditis, is the prototype for several causes of destructive thyroiditis. The precise pathophysiological mechanism of subacute thyroiditis is poorly understood; it is thought to be a postviral inflammatory process caused by common antigens in viruses and thyroid follicular cells.³² For example, thyroid follicular cells have angiotensin-converting-enzyme 2 receptors, which are recognized by SARS-CoV-2. Before the coronavirus disease 2019 pandemic, coxsackievirus and mumps virus were the viruses that were most often associated with subacute thyroiditis.

Patients with subacute thyroiditis classically present with a preceding upper respiratory illness, thyrotoxicosis, and thyroid pain that may start in one lobe and migrate to the other lobe, with radiation to the jaw or ear.³³ Fever is present in 28% of patients.³⁴ On examination, the thyroid is characteristically firm or hard, and it is tender or quite painful. Rarely, a patient with subacute thyroiditis will present without pain.

Subacute thyroiditis is frequently evident on thyroid examination, but confirmatory laboratory test results usually show evidence of inflammation (the ESR is usually >50 mm per hour, and the CRP level is elevated); in addition, transient amorphous or nodular hypoechoic areas are present on thyroid ultrasonography.³⁵ In patients with destructive thyroiditis, the ratio of total triiodothyronine (ng per deciliter) to thyroxine (μg per deciliter) is less than 20; by contrast, excess triiodothyronine relative to thyroxine is released in patients with Graves' disease or with a toxic adenoma, and the ratio is greater than 20.³⁶

DIAGNOSTIC TESTING

Dr. Grace K. Mahowald: At this hospital, the algorithm for a thyroid panel starts with evaluation of the thyrotropin level. If the thyrotropin level is less than $0.40 \mu\text{IU}$ per milliliter, then the levels of free thyroxine and total triiodothyronine are measured. This patient's thyrotropin level was low ($<0.01 \mu\text{IU}$ per milliliter; reference range, 0.40 to 5.99). The free thyroxine level was high (3.7 ng per deciliter [48 pmol per liter]; reference range, 0.9 to 1.8 ng per deciliter [12 to 23 pmol per liter]), as was the total triiodothyronine level (281 ng per deciliter [4.3 nmol per liter]; reference range, 60 to 181 ng per deciliter [0.9 to 2.8 nmol per liter]).

Antithyroid antibody tests were also performed. Thyroperoxidase antibodies can be present in patients with Hashimoto's thyroiditis or Graves' disease. The thyrotropin-receptor antibody test assesses whether antibodies that bind to the thyrotropin receptor are present (thyroid-stimulating antibodies as well as thyroid-blocking antibodies). Both the thyroperoxidase antibody test and the thyrotropin-receptor antibody test were negative.

The thyroid-stimulating immunoglobulin (TSI) test is a bioassay that is performed with cells that have been transfected with the human thyrotropin receptor and a luciferase expression construct under the control of a cyclic adenosine monophosphate promoter. The TSI index is cal-

The cause of thyrotoxicosis is traditionally determined by thyroid scintigraphy with measurement of radioactive iodine uptake over a 24-hour period.³⁷ Patients with Graves' disease and those with a toxic adenoma or toxic nodular goiter all present with a high or normal level of radioactive iodine uptake; scintigraphy shows diffuse uptake in patients with Graves' disease and focal uptake in those with toxic nodular goiter with one or more toxic adenomas. By contrast, destructive subacute thyroiditis results in thyrotoxicosis with near-absent radioactive iodine uptake. Graves' disease is readily diagnosed by means of a thyrotropin-receptor antibody test to assess thyrotropin-binding inhibition or the presence of TSI; a positive test replaces the need for measurement of radioactive iodine uptake.³⁷ In addition, the presence of vascularity on ultrasonography correlates with the level of radioactive iodine uptake.

This patient had high levels of inflammatory markers and classic physical examination findings for subacute thyroiditis. The subsequent workup included negative thyrotropin-receptor antibody tests. Reduced vascularity was observed on a bedside ultrasound image. Determination of the ratio of total triiodothyronine to thyroxine and measurement of radioactive iodine uptake were not thought to be necessary, given the classic presentation.

In patients with subacute thyroiditis, pain is managed with nonsteroidal antiinflammatory drugs, with glucocorticoid therapy reserved for

those who either do not have a response or have moderate-to-severe symptoms.³⁷ Thyrotoxicosis that develops in patients with subacute thyroiditis is caused by leakage of preformed hormone from thyroid follicles, which results in a transient hyperthyroid phase that lasts an average of 2 to 8 weeks. During the thyrotoxic phase, beta-adrenergic blockers may be used to ameliorate symptoms of hyperthyroidism, and in older patients or patients with cardiovascular disease, beta-adrenergic blockers may reduce the risk of atrial arrhythmia. When thyroid stores become exhausted and the inflammation subsides, blood levels of thyroid hormone decrease through the normal range and usually become transiently subnormal, resulting in a brief euthyroid phase followed by a variable period of hypothyroidism. After the thyrotoxicosis resolves, resumption of thyrotropin production may take a month or longer. Thus, the hypothyroid phase lasts until damaged thyroid follicular cells and the pituitary–thyroid axis recover. In this patient, hypothyroidism began approximately 6 weeks after presentation, and the patient received treatment with levothyroxine. Full recovery occurs in more than 85% of patients.³⁴

FINAL DIAGNOSIS

Subacute thyroiditis.

This case was presented at the Medicine Case Conference.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

We thank Dr. Jo-Anne O. Shepard for assistance with the radiology portion of the case presentation.

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